

# US Retail Sales and Inventory Intelligence: A Data-Driven Dashboard Approach

## 1. Project Overview and Objective

This project cleans and models a US retail transaction dataset in Excel, then builds three interactive Power BI dashboards that turn raw sales and inventory records into clear business insights.

Objective: to demonstrate end-to-end data analytics — from data pre-processing in Excel to an interactive, multi-page Power BI report — so that a store manager or business owner can quickly see sales performance, product/category trends, and inventory health, and filter the data themselves without needing to write any code.

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## 2. Data Sources

|                              |                                                                                                                 |
|------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Source file</b>           | US_Retail_Sales_Cleaned_Data.xlsx (workbook prepared in Excel)                                                  |
| <b>Domain</b>                | Retail / Inventory & Sales Analytics                                                                            |
| <b>Timeline</b>              | Jan 2024 – Jun 2025 (transaction dates)                                                                         |
| <b>Raw rows → Clean rows</b> | 1,145 raw rows → 1,082 cleaned rows in Fact_Sales (63 exact duplicates removed)                                 |
| <b>Structure</b>             | 1 fact table (Fact_Sales) + 3 dimension tables (Dim_Date, Dim_Store, Dim_Product) + 4 pre-built pivot summaries |

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## 3. Problem Statement

- To analyse sales performance across stores/regions and time periods to identify top-performing locations and seasonal trends.
  - To study which product categories and individual products drive the most revenue and quantity sold.
  - To evaluate stock levels against reorder points and flag products at risk of stock-out.
  - To understand customer behaviour (New / Returning / Member) and preferred payment methods, and measure the effect of promotions on sales.
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#### 4. Attribute (Column /Features) Details:

Fact\_Sales (transaction-level fact table):

| Attribute Name    | Data Type            | Description                                                             |
|-------------------|----------------------|-------------------------------------------------------------------------|
| Transaction_ID    | Text (Unique ID)     | Unique identifier for each sales transaction                            |
| Date              | Date                 | Date the transaction occurred                                           |
| Store_ID          | Text (Categorical)   | Links to Dim_Store (which store made the sale)                          |
| Product_ID        | Text (Categorical)   | Links to Dim_Product (which item was sold)                              |
| Customer_ID       | Text                 | Unique identifier for the customer                                      |
| Customer_Type     | Categorical          | New / Returning / Member                                                |
| Payment_Method    | Categorical          | Cash / Credit Card / Debit Card / Mobile Pay                            |
| Promotion_Applied | Categorical (Yes/No) | Whether a promotion/discount applied to the sale                        |
| Quantity_Sold     | Numeric (Integer)    | Units sold in the transaction                                           |
| Unit_Price        | Numeric (Currency)   | Price per unit (\$)                                                     |
| Total_Sales       | Numeric (Currency)   | $\text{Quantity\_Sold} \times \text{Unit\_Price}$                       |
| Stock_Available   | Numeric (Integer)    | Units left in stock at time of sale                                     |
| Reorder_Level     | Numeric (Integer)    | Stock threshold that should trigger reordering                          |
| Stock_Status      | Categorical          | 'Adequate' or 'Low' — Stock_Available compared with Reorder_Level       |
| Product_Category  | Categorical          | Grocery, Electronics, Snacks, Beverages, Personal Care, Home Essentials |
| Store_Location    | Text                 | City, State of the store                                                |

Dim\_Date, Dim\_Store, Dim\_Product: standard lookup tables — Dim\_Date breaks each date into Year, Quarter, Month\_Name and Weekday; Dim\_Store maps Store\_ID to City/State; Dim\_Product maps Product\_ID to Product\_Name/Category.

## 5. Tools & Technologies

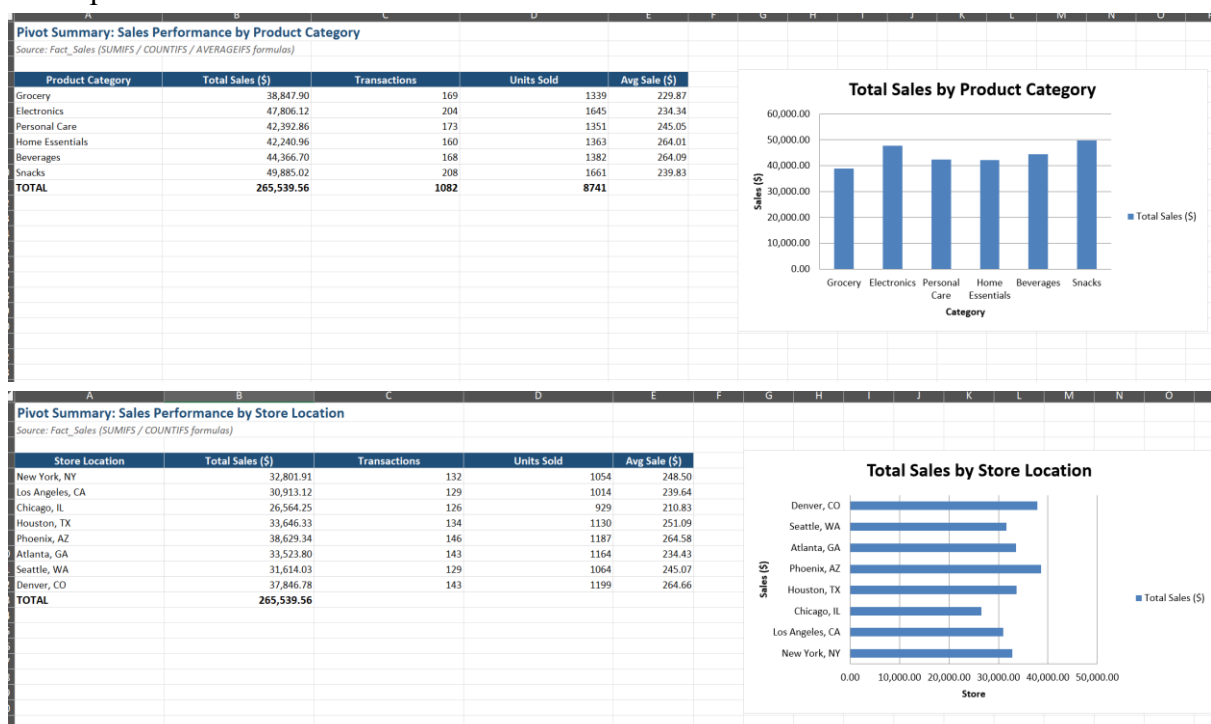
- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and 3 interactive dashboards built in this Report.

## 6. Data Pre-Processing (Excel / Power Query)

### Tasks Performed:

The workbook's 'Cleaning Log' sheet records every step already performed on the raw 1,145-row extract:

- Removed 63 fully duplicated rows → 1,082 rows remain.
- Standardised text casing/whitespace across Store ID, Product ID, Category, Customer Type, Payment Method, Promotion Applied and Store Location (e.g. 'ELECTRONICS' / 'electronics' → 'Electronics').
- Filled 32 missing Store Location values using a Store ID lookup.
- Parsed dates from 5 mixed formats into one standard date field.
- Imputed missing Quantity Sold, Stock Available and Reorder Level using the median for that product/category (22, 43 and 32 rows respectively).
- Filled missing Customer Type / Payment Method with 'Unknown' rather than deleting the row.
- Added calculated fields: Total\_Sales, Stock\_Status, Year, Month\_Number, Month\_Name, Quarter, Weekday.
- Split the flat table into 1 Fact table + 3 Dimension tables, ready for a star-schema import into Power BI.





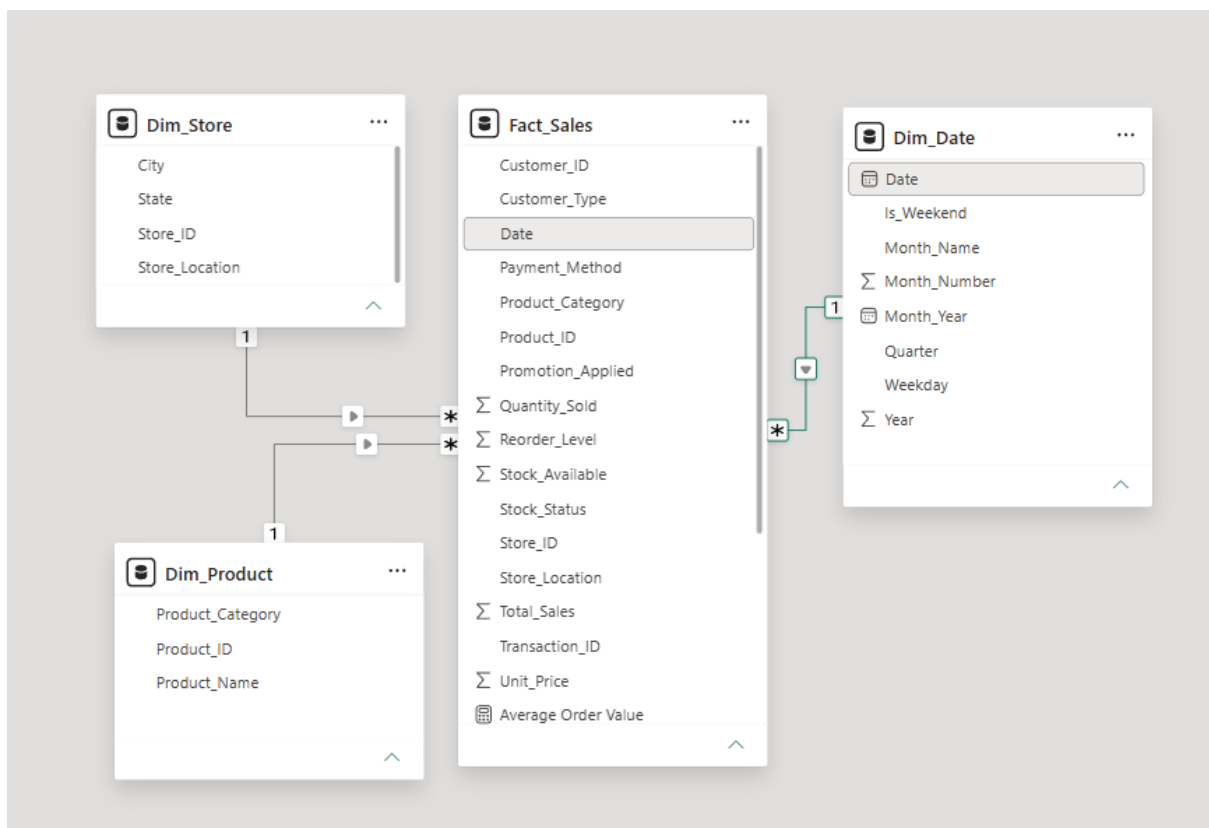
- Click 'Transform Data' to open Power Query, confirm each table's first row is used as headers and each column has the correct data type (Date column = Date, Total\_Sales = Decimal Number, etc.), then click 'Close & Apply'.

## 7.2 Building the Data Model (Relationships)

Go to the Model view (the icon on the left-hand strip that looks like three connected boxes). Power BI usually auto-detects these, but check/create them manually:

|                                                                 |                               |
|-----------------------------------------------------------------|-------------------------------|
| <b>Fact_Sales[Date] →<br/>Dim_Date[Date]</b>                    | Many-to-one, single direction |
| <b>Fact_Sales[Store_ID] →<br/>Dim_Store[Store_ID]</b>           | Many-to-one, single direction |
| <b>Fact_Sales[Product_ID]<br/>→<br/>Dim_Product[Product_ID]</b> | Many-to-one, single direction |

To create one: click a field in the fact table and drag it onto the matching field in the dimension table; a relationship line appears. This 'star schema' (one fact table in the middle, dimension tables around it) is what lets slicers on Store or Product filter every chart correctly.



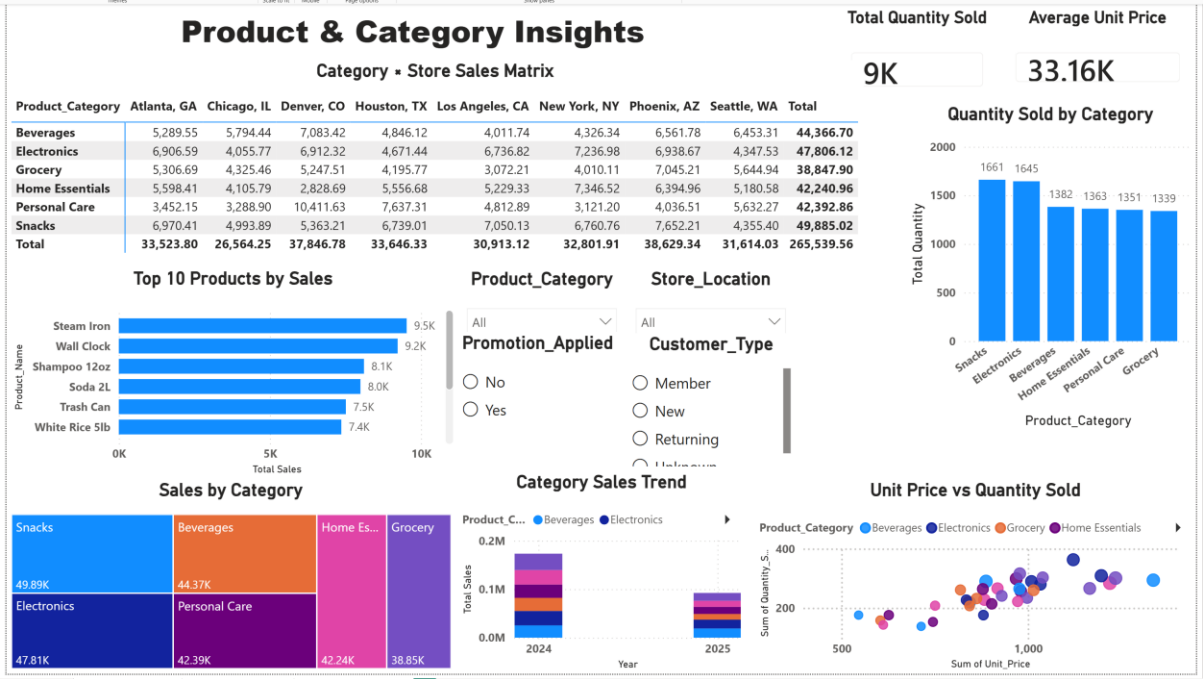
### 7.3 DAX Measures Used

Go to the Report view → select Fact\_Sales in the Fields pane → Home ribbon (or right-click Fact\_Sales) → 'New Measure', then type each formula below and press Enter.

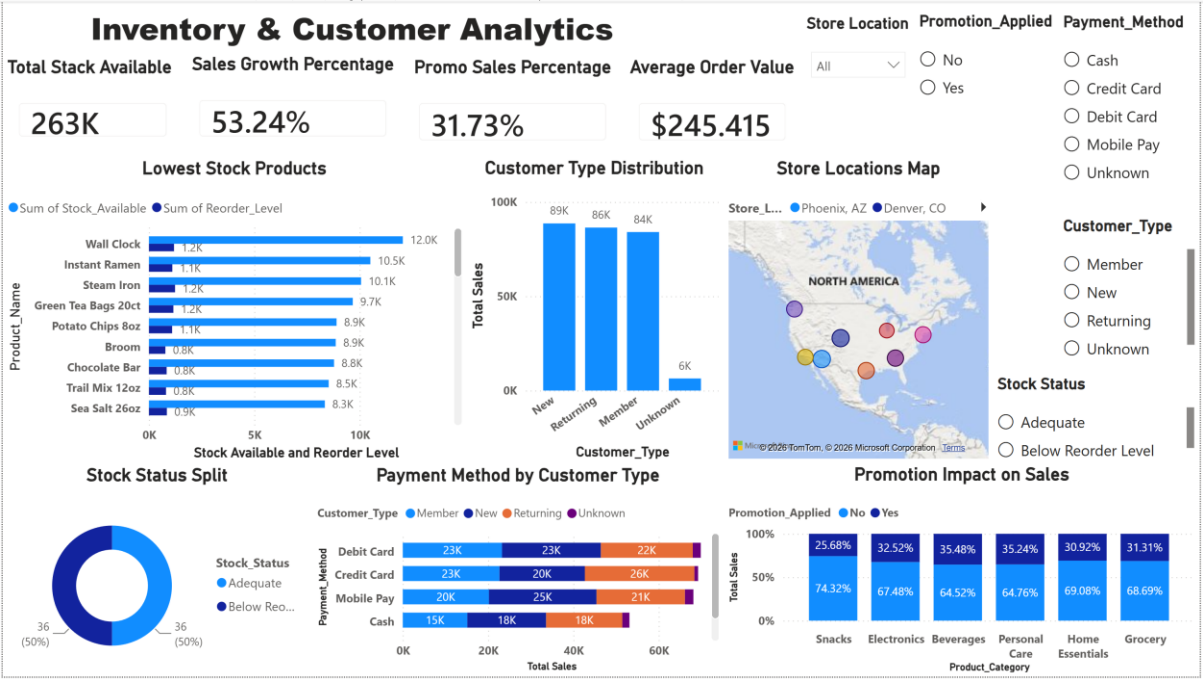
| Measure               | DAX Formula                                                                         | Purpose                                             |
|-----------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------|
| Total Sales           | SUM(Fact_Sales[Total_Sales])                                                        | Overall revenue KPI and base of most charts         |
| Total Quantity        | SUM(Fact_Sales[Quantity_Sold])                                                      | Units-sold KPI                                      |
| Total Transactions    | DISTINCTCOUNT(Fact_Sales[Transaction_ID])                                           | Transaction-count KPI                               |
| Average Order Value   | DIVIDE([Total Sales],[Total Transactions])                                          | Sales ÷ Transactions , avoids divide-by-zero errors |
| Total Stock Available | SUM(Fact_Sales[Stock_Available])                                                    | Inventory KPI                                       |
| Low Stock Items       | CALCULATE(DISTINCTCOUNT(Fact_Sales[Product_ID]), Fact_Sales[Stock_Status]="Low")    | Reorder-alert KPI / gauge                           |
| Sales LY              | CALCULATE([Total Sales], SAMEPERIODLASTYEAR(Dim_Date[Date]))                        | Year-over-year comparison line                      |
| Sales Growth %        | DIVIDE([Total Sales]-[Sales LY],[Sales LY])                                         | Growth trend / gauge target check                   |
| Promo Sales %         | DIVIDE(CALCULATE([Total Sales], Fact_Sales[Promotion_Applied]="Yes"),[Total Sales]) | Share of revenue from promotions                    |



Dashboard 2 — Product & Category Insights



Dashboard 3 — Inventory & Customer Analytics



## 9. Insights & Conclusions

### Key Findings

- Sales are spread across 8 stores and 6 categories, with clear month-to-month and weekday variation visible in the trend and ribbon charts.
- A small set of products and one or two categories account for a disproportionate share of revenue — visible in the Top 10 Products bar and the Treemap.
- A subset of products consistently run at or below their reorder level, creating stock-out risk that the Reorder Alert bookmark surfaces immediately.
- Promotions and payment method both shift the customer mix — the 100% stacked column and payment-method charts make this comparison direct.

### Analysis Insights

|                     |                                                                                                                                                               |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Descriptive</b>  | What happened — the KPI cards and trend/column charts summarise total sales, quantity, and store/category performance for the period.                         |
| <b>Diagnostic</b>   | Why it happened — the scatter chart (price vs quantity) and the promotion/payment-method breakdowns explain drivers behind the sales pattern.                 |
| <b>Predictive</b>   | What could happen next — the monthly trend line and Year-over-Year growth measure (Sales Growth %) hint at where revenue is heading if the pattern continues. |
| <b>Prescriptive</b> | What to do about it — the Reorder Alert bookmark and Low Stock chart point straight to the products that need restocking now.                                 |

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## 10. Conclusions

Combining Excel for cleaning/pre-processing with Power BI for modelling and visualisation turned a raw, inconsistent 1,145-row export into a reliable, interactive 3-dashboard report. The star-schema data model, DAX measures, slicers, tooltips, bookmarks and drill-downs together let anyone — even without any technical background — explore sales performance, product/category trends, and inventory risk on their own, and switch views instantly using the bookmark buttons.